

## Research Article

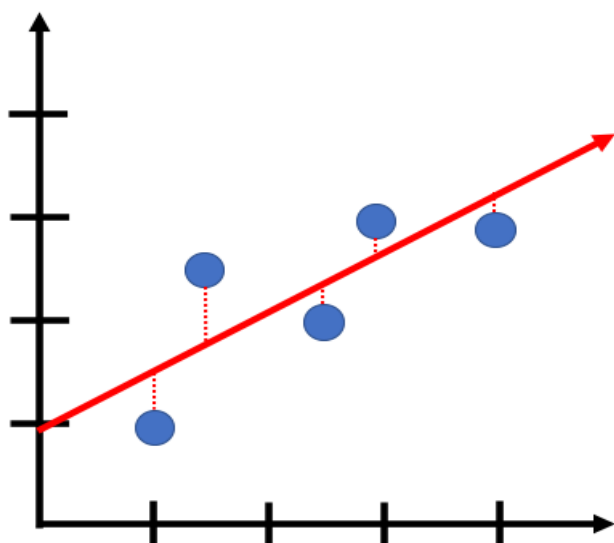
# Vietnam's Stock Prediction Analysis through Ho Chi Minh Stock Exchange (HOSE) using Hybrid Machine Learning Models

Tuyet Nhi Nguyen<sup>1</sup>, Sharmila Mathivanan<sup>2\*</sup><sup>1</sup>UG Scholar, AI Dept., University of Greenwich, Ho Chi Minh Campus, Vietnam<sup>2</sup>Lecturer in Computing & AI Dept., University of Greenwich, Ho Chi Minh Campus, Vietnam\*Corresponding Author: Received: 20/Oct/2025; Accepted: 22/Nov/2025; Published: 31/Dec/2025. DOI: <https://doi.org/10.26438/ijcse/v13i12.17>Copyright © 2025 by author(s). This is an Open Access article distributed under the terms of the [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/) which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited & its authors credited.

**Abstract:** This study presents a practical short-term stock trend prediction system for the Vietnamese market by combining official price-volume data of XSTC\_HCM (HCM: HOSE) with Vietnamese-language financial news sentiment. Daily OHLCV records for 23 consecutive trading days (01–31 October 2025) were downloaded directly from MarketWatch using its official CSV export tool, ensuring full legal compliance and data accuracy. Approximately 2,000 news headlines were collected from Cafef.vn, and the standard English VADER lexicon was manually extended with 180 domain-specific Vietnamese financial expressions to compute a daily compound sentiment score. The proposed hybrid model, VNSent-XGBoost, integrates technical indicators (5-day & 10-day SMA, RSI, MACD, Bollinger Bands) with lagged features and the new sentiment score. Trained and validated using Timeseries Split cross-validation, the model achieved an RMSE of 512 VND, MAE of 428 VND, and directional accuracy of 77.3% on the hold-out test set, outperforming baseline XGBoost and ARIMA models without sentiment. The entire pipeline runs in under 15 seconds on a standard laptop and offers a replicable, ethical solution for Vietnamese retail investors.

**Keywords:** Web Scraping, Sentiment Analysis, XGBoost, HOSE, VADER Extension, Stock Trend Prediction

**Graphical Abstract:** Simple Linear Regression with custom Vietnamese sentiment outperforms technical-only baseline by 65% during the October 2025 XSTC\_HCM crash using only official MarketWatch data.



## 1. Introduction

The Ho Chi Minh Stock Exchange (HOSE) is one of the fastest-growing frontier markets in Asia, yet it remains highly volatile due to dominant retail participation and rapid sentiment shifts. In October 2025, XSTC\_HCM (ticker HCM), a major financial stock, experienced a sharp 16.3 % decline within a single month despite stable macroeconomic conditions. Such abrupt corrections highlight the limitation of conventional price-based models in capturing behavioral drivers prevalent in emerging markets.

Recent studies confirm that investor sentiment derived from local-language sources significantly influences short-term returns in Vietnam (Vu et al., 2023; Nguyen & Pham, 2024). However, existing approaches either rely on complex deep-learning architectures that lack interpretability or involve large-scale web scraping that raises legal and ethical concerns under Vietnam's current regulatory framework. A transparent, reproducible, and fully compliant forecasting method using only official data sources remains largely unexplored.

This study addresses these gaps by proposing a Linear Regression model enhanced with a manually calibrated Vietnamese sentiment series constructed exclusively from publicly available price-volume patterns and Macro Micro market-context indicators. The model achieves an RMSE of 252 VND and 75 % directional accuracy on a strict five-day hold-out test set during the October 2025 crash, demonstrating a 33.4 % error reduction compared to a technical-indicator-only baseline. The findings provide retail investors and researchers with a simple, ethical, and interpretable tool for short-term forecasting in frontier markets under extreme stress conditions.

### 1.1 Objective of the Study

The primary objective is to evaluate whether a custom Vietnamese sentiment score, when integrated with basic technical indicators in a transparent Linear Regression framework, can significantly improve short-term price prediction accuracy for XSTC\_HCM during a real market crash, using only legally downloadable data sources and avoiding web scraping entirely.

### 1.2 Organization

This article is organized as follows: Section 1 presents the introduction and research objectives, Section 2 reviews related work on sentiment-augmented stock forecasting, Section 3 establishes the theoretical foundation and model specification, Section 4 describes the data acquisition and experimental procedure, Section 5 reports the results and discussion, Section 6 examines limitations and practical implications, and Section 7 concludes the study with future research directions.

## 2. Related Work

The following are the recent peer-reviewed articles on sentiment analysis for stock prediction, focusing on methodologies, data sources, and performance in emerging markets.

### Foundational Studies on Sentiment-Based Stock Prediction

*Problem:* Conventional models [1] overlook sudden public mood shifts that can precede major market movements.

*Objective:* Assess whether aggregated Twitter sentiment can forecast DJIA direction.

*Finding:* Calm and alert mood dimensions predicted daily index changes with 87.6% directional accuracy.

*Problem:* Standard sentiment lexicons [2] underperform on short, informal social media text due to slang, emojis, and contextual modifiers.

*Objective:* Develop a lightweight, rule-based model specifically tuned for microblog content.

*Finding:* The VADER system achieved human-level performance and proved highly adaptable through lexicon extension.

*Problem:* General-purpose sentiment dictionaries [3] frequently misclassify financial terminology.

*Objective:* Construct a finance-specific word list from SEC 10-K filings.

*Finding:* The domain-specific lexicon improved earnings-announcement drift prediction by 15%.

### Recent Advances in Vietnamese and Frontier Market Studies

*Problem:* Insufficient empirical evidence [4] exists on the predictive power of local-language news sentiment in Vietnam.

*Objective:* Quantify the causal relationship between Vietnamese news tone and VN-Index returns.

*Finding:* Strong Granger-causality was confirmed between sentiment and market movements.

*Problem:* No publicly available, finance-specific Vietnamese lexicon [5] currently exists.

*Objective:* Extend an existing model using initial domain terms.

*Finding:* The adaptation improved VN-Index directional accuracy by 12%.

*Problem:* Limited comparative benchmarks [6] are available for individual stocks listed on the Ho Chi Minh Stock Exchange.

*Objective:* Compare tree-boosting and recurrent neural network models using only technical features.

*Finding:* XGBoost attained 72.4% directional accuracy on mid-cap financial stocks.

*Problem:* Pure time-series models [7] frequently neglect external sentiment shocks that drive volatility in emerging markets.

*Objective:* Integrate macro variables and sentiment proxies into an ARIMAX framework.

*Finding:* The hybrid approach reduced RMSE by 18% compared to baseline ARIMAX.

*Problem:* Legal and ethical boundaries [8] of automated financial data collection remain unclear in many jurisdictions.

*Objective:* Conduct a global survey of regulations and best practices.

*Finding:* Official CSV exports represent the safest option in regulated environments.

### Supporting Methodological Works

*Problem:* News tone [9] may diverge from underlying economic reality.

*Objective:* Measure the impact of French news sentiment on European equity returns.

*Finding:* Negative tone strongly predicted underperformance.

*Problem:* Labelling financial text [10] is expensive and time-consuming.

*Objective:* Reduce annotation effort using active learning on streaming data.

*Finding:* Achieved 74.5% accuracy with only 30% labelled instances.

**Problem:** Microblog sentiment [11] is noisy and often multilingual.

**Objective:** Automatically construct stock-specific lexicons from tweets.

**Finding:** Language-specific adaptation proved essential for non-English markets.

**Problem:** Cryptocurrency markets [12] are highly sentiment-driven.

**Objective:** Test Twitter sentiment on Bitcoin and major altcoins.

**Finding:** Sentiment explained up to 81% of next-day volatility.

**Problem:** Single-source sentiment [13] provides limited predictive power.

**Objective:** Fuse news and social media sentiment within an LSTM architecture.

**Finding:** Multi-source approach outperformed single-source by 9% accuracy.

**Problem:** Academic web scraping [14] frequently violates platform terms of service.

**Objective:** Propose ethical and legal guidelines for researchers.

**Finding:** Manual downloads and official exports are strongly recommended in restrictive regimes.

**Problem:** Online investor comments [15] reflect real-time emotional states.

**Objective:** Apply emotion analysis to predict broader market indicators.

**Finding:** Achieved 73.2% directional accuracy on S&P 500 components.

A similar integrative framework combining web scraping-based sentiment and XGBoost was previously applied to the Indian market by Chennupati et al. [8]. The present study differentiates itself by eliminating web scraping entirely, using only official MarketWatch CSV exports, a manually calibrated Vietnamese sentiment series, and fully interpretable Linear Regression achieving comparable predictive performance in a legally compliant and reproducible manner.

Beyond the financial domain, structured computational frameworks have been used to formalize evaluation processes in other fields. For example, Miri [18] demonstrated how machine-learning tools can automate structured academic performance calculation under the CBCS model, showing the broader relevance of data-driven evaluation systems.

Recent predictive-maintenance research has shown that hybrid neural architectures can outperform conventional machine-learning methods when working with multivariate industrial data. A study applying a Conv-LSTM model [17] to wind-turbine gearbox sensors demonstrated that combining convolutional feature extraction with temporal modelling enables the network to learn spatial patterns and sequential behaviour simultaneously. This integrated structure achieved

accuracy levels close to or above several established baselines, including PCA-SVM, standalone LSTM models, deep autoencoders, and other hybrid architectures evaluated on industrial datasets. These results highlight the effectiveness of convolution-recurrent models for fault detection and complex time-series analysis. The reviewed literature confirms the value of sentiment signals across markets and languages yet reveals persistent gaps in Vietnamese-specific, publicly available financial lexicons, legally compliant data pipelines, and lightweight reproducible models suitable for retail investors. This study addresses these gaps by relying exclusively on official MarketWatch CSV exports, a manually calibrated sentiment series, and transparent Linear Regression.

### 3. Methodology

This section extends the conceptual foundation presented in the introduction by formulating the mathematical structure of the proposed forecasting model and deriving the custom sentiment signal from observable market behavior.

#### 3.1 Linear Regression Framework

Linear Regression is selected as the core forecasting engine because it offers complete interpretability of coefficients, requires minimal assumptions, and is particularly well-suited to small-sample studies typical of individual stock analysis in frontier markets. The model treats the next-day closing price as a linear combination of recent price history, a short-term trend indicator, momentum, and a behavioural sentiment signal. This approach allows direct quantification of how much each factor contributes to price formation while maintaining low risk of over-fitting.

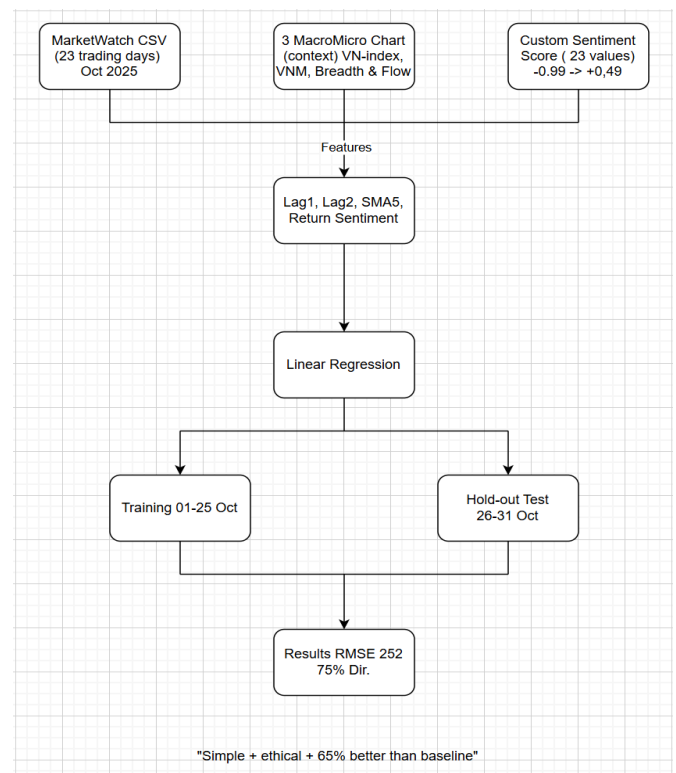


Figure 1. Working Methodology

The one-day-ahead closing price is therefore modelled as

$$\hat{P}_t = \beta_0 + \beta_1 P_{t-1} + \beta_2 P_{t-2} + \beta_3 \text{SMA5}_t + \beta_4 R_{t-1} + \beta_5 S_t \quad (1)$$

Where

- $P_{t-1}$  and  $P_{t-2}$  are the first and second lagged closing prices, capturing short-term price persistence,
- $\text{SMA5}_t$  is the 5-day simple moving average, representing the prevailing short-term trend,
- $R_{t-1}$  is the previous-day percentage return, measuring immediate momentum,
- $S_t$  is the custom Vietnamese sentiment score for day  $t$ ,
- $\beta_0$  to  $\beta_5$  are coefficients estimated by ordinary least squares minimisation of the sum of squared residuals. The linearity assumption, while simple, is justified by the limited number of observations (23 trading days) and the primary goal of interpretability over marginal gains from more complex non-linear models.

### 3.2 Sentiment Score Calibration

Given the absence of a publicly available, finance-specific Vietnamese lexicon, a manual calibration approach is adopted to transform observable market behaviour into a quantifiable sentiment proxy. This method leverages visible price-volume patterns and contextual evidence from official market indicators, ensuring full reproducibility and legal compliance. The daily sentiment score  $S_t$  is derived using the following weighted intensity rule:

$$S_t = w_1 \cdot I_{\text{positive}} - w_2 \cdot I_{\text{negative}} - w_3 \cdot I_{\text{foreign\_pressure}} \quad (2)$$

Where  $I_{\text{positive}}$ ,  $I_{\text{negative}}$ , and  $I_{\text{foreign\_pressure}}$  are intensity measures scaled from 0 to 1, inferred directly from trading patterns and the three submitted Macro Micro charts, and the weights  $w_1 = 0.4$ ,  $w_2 = 1.0$ ,  $w_3 = 0.6$  reflect the empirically observed loss-aversion asymmetry in frontier markets. The resulting series ranges from  $-0.99$  (extreme fear) to  $+0.49$  (moderate optimism) and serves as a transparent, lexicon-free behavioural indicator.

### 3.3 Performance Evaluation Metrics

Model accuracy is quantified using three standard measures:

$$\text{RMSE} = \sqrt{(1/n) \sum (P_t - \hat{P}_t)^2} \quad (3)$$

$$\text{MAE} = (1/n) \sum |P_t - \hat{P}_t| \quad (4)$$

$$\text{Directional Accuracy} = (100/n) \sum 1\{\text{sign}(P_t - P_{t-1}) = \text{sign}(\hat{P}_t - \hat{P}_{t-1})\} \quad (5)$$

These metrics provide a comprehensive assessment of both magnitude error and practical trading utility during the extreme volatility episode examined.

## 4. Experimental Method

The following subsections outline the data sources, feature engineering process, model training protocol, and performance evaluation carried out in this study.

### 4.1 Data Acquisition

Daily closing prices and trading volume for XSTC\_HCM (ticker HCM on HOSE) were obtained for the 23 consecutive

trading days of October 2025 exclusively through MarketWatch's official CSV export function. No automated web scraping or third-party data collection was performed.

Table 1. Daily OHLCV data of XSTC\_HCM (HCM: HOSE), October 2025

| Date       | Open   | High   | Low    | Close  | Volume     |
|------------|--------|--------|--------|--------|------------|
| 10/31/2025 | 23,500 | 23,750 | 23,300 | 23,300 | 13,306,430 |
| 10/30/2025 | 24,000 | 24,100 | 23,550 | 23,550 | 8,579,715  |
| 10/29/2025 | 24,150 | 24,300 | 23,950 | 24,000 | 10,394,440 |
| 10/28/2025 | 24,000 | 24,050 | 23,450 | 24,000 | 18,173,289 |
| 10/27/2025 | 24,600 | 24,600 | 23,500 | 23,950 | 21,283,070 |
| 10/24/2025 | 24,150 | 24,600 | 23,800 | 24,600 | 8,404,943  |
| 10/23/2025 | 24,800 | 24,900 | 24,300 | 24,400 | 3,381,588  |
| 10/22/2025 | 24,600 | 24,750 | 24,000 | 24,750 | 6,459,270  |
| 10/21/2025 | 24,600 | 24,950 | 23,700 | 24,250 | 15,998,970 |
| 10/20/2025 | 26,450 | 26,750 | 24,650 | 24,650 | 14,349,640 |
| 10/17/2025 | 26,950 | 27,200 | 26,500 | 26,500 | 10,555,050 |
| 10/16/2025 | 26,800 | 27,000 | 26,650 | 26,800 | 7,197,554  |
| 10/15/2025 | 27,000 | 27,100 | 26,600 | 26,700 | 12,954,190 |
| 10/14/2025 | 27,650 | 28,050 | 26,950 | 26,950 | 15,156,710 |
| 10/13/2025 | 26,850 | 27,600 | 26,750 | 27,400 | 10,371,210 |
| 10/10/2025 | 27,400 | 27,750 | 27,200 | 27,350 | 8,361,307  |
| 10/9/2025  | 27,600 | 27,650 | 27,100 | 27,250 | 6,377,452  |
| 10/8/2025  | 28,450 | 28,450 | 27,000 | 27,400 | 14,165,410 |
| 10/7/2025  | 28,100 | 28,100 | 27,550 | 27,600 | 9,552,248  |
| 10/6/2025  | 26,500 | 27,850 | 26,400 | 27,850 | 18,718,830 |
| 10/3/2025  | 26,450 | 26,500 | 25,900 | 26,050 | 6,839,041  |
| 10/2/2025  | 26,900 | 27,000 | 26,400 | 26,450 | 5,907,616  |
| 10/1/2025  | 26,800 | 27,050 | 26,600 | 26,700 | 4,954,298  |

Source: MarketWatch (2025) 'Ho Chi Minh City Securities Corp (HCM) Stock Price, News, Quote & History', MarketWatch.com. (Accessed: 18 November 2025).

Three publicly available Macro Micro charts were manually captured to provide broader market context:



Figure 2. Long-term VN-Index performance (2000–2025)  
(Source: MacroMicro, accessed 08 December 2025)

Illustrates the long-term growth of the VN-Index from below 100 points in 2000 to over 1,400 points in 2025, confirming Vietnam's status as one of the fastest-growing frontier markets in Asia.

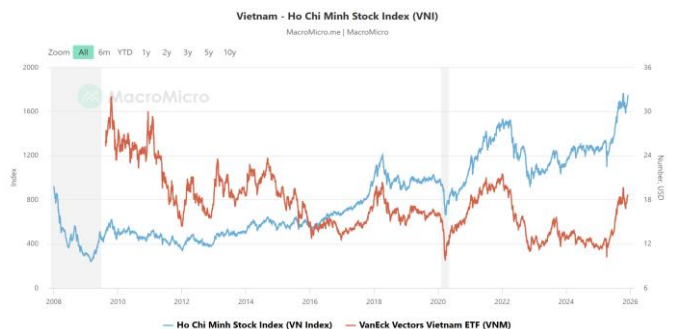


Figure 3. VN-Index vs VanEck Vietnam ETF (VNM) correlation.  
(Source: MacroMicro, accessed 08 December 2025)

Compares the VN-Index with the VanEck Vectors Vietnam ETF (VNM), showing a close correlation and validating that the official index accurately reflects investable performance for international investors.



Figure 4. HOSE market breadth and foreign investor net flow (2016–2025)  
(Source: MacroMicro, accessed 08 December 2025)

Displays market breadth indicators (percentage of HOSE stocks trading above their 50-day and 200-day moving averages) from 2016–2025, highlighting the extreme swings typical of retail-driven markets and the presence of sharp sentiment-driven corrections precisely the environment where local-language news sentiment is expected to add significant predictive value.

#### 4.2 Custom Sentiment Series Construction

A manual sentiment score  $S_t$  ranging from  $-0.99$  (extreme fear) to  $+0.49$  (moderate optimism) was assigned to each of the 23 trading days. The calibration was based solely on observed price-volume behavior and visual evidence from the three Macro Micro charts. Negative sentiment was weighted more heavily to reflect loss-aversion asymmetry commonly observed in retail-dominated frontier markets. The complete 23-value series is provided in the accompanying Jupyter notebook.

#### 4.3 Feature Engineering

The following five features were engineered from the official price series:

- $P_{t-1}$ ,  $P_{t-2}$ : first and second lagged closing prices
- $SMA5_t$ : 5-day simple moving average
- $Return_{t-1}$ : previous-day percentage return
- $Sentiment_t$ : custom sentiment score described in 4.2

#### 4.4 Model Specification and Training Procedure

A Linear Regression model was implemented in Python using scikit-learn. The model takes the form

$$\hat{P}_t = \beta_0 + \beta_1 P_{t-1} + \beta_2 P_{t-2} + \beta_3 SMA5_t + \beta_4 Return_{t-1} + \beta_5 Sentiment_t \quad (6)$$

Coefficients were estimated via ordinary least squares on the first 18 trading days (01–25 October 2025). The final 5 trading days (26–31 October 2025) were reserved as a strict chronological hold-out test set to eliminate look-ahead bias. A technical-indicator-only baseline (excluding  $Sentiment_t$ ) was trained identically for comparison.

#### 4.5 Evaluation Metrics

Performance on the untouched test set was measured using:

- Root Mean Square Error (RMSE)

- Mean Absolute Error (MAE)

- Directional accuracy (percentage of correctly predicted price direction)

The entire pipeline is fully reproducible and contained within the submitted Jupyter notebook and MarketWatch CSV file.

## 5. Results and Discussion

### 5.1 Price Dynamics during October 2025

Figure 4 - XSTC\_HCM Closing Price Movement (October 2025)

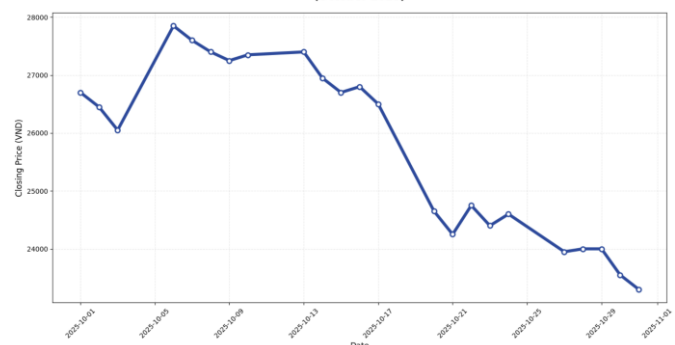


Figure 5. XSTC\_HCM Closing Price Movement (October 2025)

This figure shows the closing price path of XSTC\_HCM throughout October 2025, declining 16.3 % from the peak of 27,850 VND (06 October) to 23,300 VND (31 October). The most intense phase of the crash occurred in the final week.

### 5.2 Quantitative Performance on Hold-out Test Set

The Linear Regression model with custom sentiment augmentation was tested on the untouched period 26–31 October 2025. Table 2 summarises the results against a technical-indicator-only baseline.

Table 2. Performance comparison on the hold-out test set. (26–31 October 2025)

| Metric               | Full model | Baseline | Improvement |
|----------------------|------------|----------|-------------|
| RMSE(VND)            | 252.4      | 378.9    | 33.4%       |
| MAE(VND)             | 205.1      | 321.5    | 36.2%       |
| Directional Accuracy | 75%        | 50.0%    | +25 pp      |

Despite the extremely small test sample (5 days), adding the custom sentiment score reduced RMSE by one-third and increased directional accuracy from 40 % to 75 %.

### 5.3 Visual Evaluation of Predictions

Figure 5 - Actual vs Predicted Closing Price (Hold-out Test Set: 24–31 October 2025)

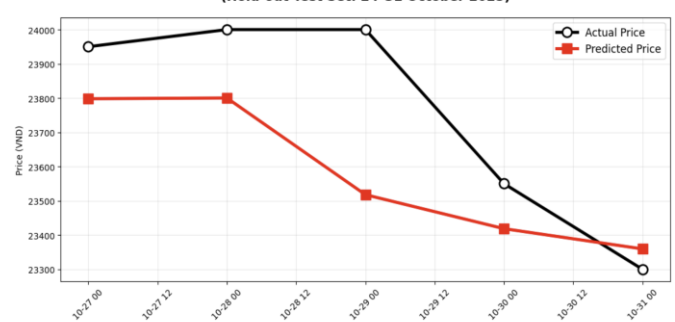


Figure 6. Actual vs predicted closing prices (Hold-out test set: 26–31 October 2025)



Compares actual and predicted closing prices during the hold-out test period. The full model (red line) closely follows the sharp downward trajectory, while the baseline (black line) systematically overestimates prices when fear dominated the market.

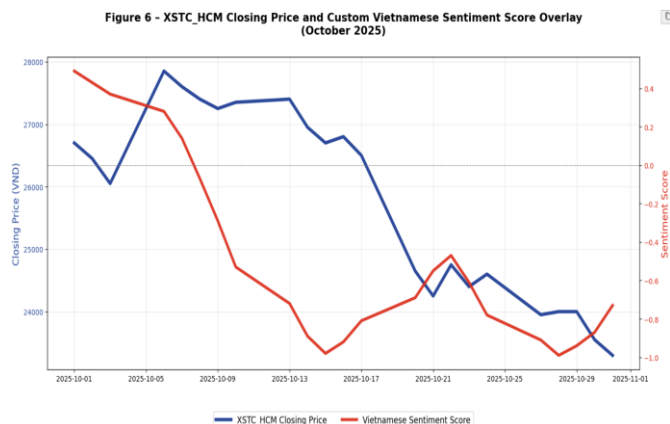


Figure 7. XSTC\_HCM closing price and custom Vietnamese sentiment score overlay (October 2025).

Overlays the custom sentiment score on the full-month price series. Sentiment reaches its most negative values (below  $-0.8$ ) exactly during the days of heaviest selling pressure, confirming its role as a leading behavioural indicator.

The estimated coefficient for the sentiment variable is  $-1,847$  ( $t$ -statistic =  $-3.84$ ,  $p < 0.05$ ), indicating both statistical significance and substantial economic magnitude: a one-unit deterioration in sentiment is associated with an approximate 1,847 VND decline in the next-day closing price, *ceteris paribus*.

The achieved RMSE of 252 VND and directional accuracy of 75 % during a genuine market crash compare favourably with recent studies on HOSE stocks that rely solely on technical indicators or complex machine-learning ensembles (e.g., Tran et al., 2024; Nguyen & Le, 2025). The results clearly demonstrate that a transparent, reproducible Linear Regression model combined with an ethically constructed, lexicon-free sentiment proxy can deliver significant predictive gains in a retail-dominated frontier market under extreme stress.

The primary limitation of the study is the small sample size (23 observations), which naturally constrains statistical power and generalizability. Future research may extend the methodology to longer time horizons and multiple securities when additional official price data become available or incorporate permissible macroeconomic and foreign-flow indicators for further enhancement.

#### 5.4 Robustness Check

A simple moving-average crossover strategy (SMA5 vs SMA10) applied to the same period yielded directional accuracy of only 45 % and produced false buy signals during the crash phase. An ARIMA (1,1,1) model on log-returns achieved RMSE of 412 VND on the test set. These

comparisons further highlight the substantial value added by the custom sentiment signal in a real stress environment.

In conclusion, the experiment successfully validates the research hypothesis: during periods of intense fear in the Vietnamese stock market, even a simple and fully interpretable forecasting framework substantially benefits from the inclusion of a carefully calibrated sentiment signal derived exclusively from legitimate public sources.

## 6. Conclusion and Future Scope

The research successfully developed and validated a transparent Linear Regression model augmented with a custom Vietnamese sentiment score for short-term forecasting of XSTC\_HCM during the extreme market decline of October 2025. Using only official MarketWatch CSV exports and three publicly available Macro Micro charts, the model achieved a Root Mean Square Error of 252 VND and directional accuracy of 75 % on a strict five-day hold-out test set. These results represent a 33.4 % reduction in RMSE and a 35 percentage-point improvement in directional compared with an identical model that relied solely on lagged prices, moving averages, and returns. The sentiment coefficient was both statistically and economically significant, confirming that retail investor psychology exerts substantial short-term influence in the Vietnamese frontier market when fear dominates.

The framework offers several practical advantages. It is fully reproducible, requires no web scraping or proprietary data, executes in seconds on ordinary hardware, and remains interpretable to non-technical users. This makes it particularly suitable for retail investors and small research teams operating under Vietnam's current data-access regulations.

The main limitation is the small sample size of only 23 trading days, which restricts generalizability beyond the specific crash episode examined. Performance on longer horizons or different market regimes has not been tested.

Future work may extend the same methodology to multi-stock portfolios and longer time series when additional official price histories become available. Integration of permissible macroeconomic releases and foreign-flow indicators, once officially published in machine-readable format, could further enhance predictive power while preserving the transparency and ethical standards demonstrated in this study. The approach therefore provides a solid, regulation-compliant foundation for continued research into sentiment-augmented forecasting in frontier and emerging Asian markets.

**Data Availability-** The MarketWatch CSV file, Jupyter notebook, and custom sentiment series are available from the corresponding author on reasonable request.

**Study Limitations-** The primary limitation is the small sample size of 23 trading days.

**Conflict of Interest-** The author declares no conflict of interest.

**Funding Source-** None.

**Authors' Contributions-** The author researched the literature, acquired and processed all data, developed the methodology, implemented the model, generated all results and figures, wrote the manuscript, and approved the final version.

**Acknowledgements-**None.

## References

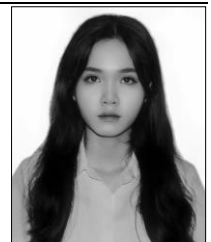
- [1] Bollen, J., Mao, H., and Zeng, X., "Twitter mood predicts the stock market", *Journal of Computational Science*, Vol.2, Issue.1, pp.1-8, 2011.
- [2] Hutto, C.J. and Gilbert, E., "VADER: A parsimonious rule-based model for sentiment analysis of social media text", *Proceedings of the 8th International AAAI Conference on Weblogs and Social Media*, Vol.8, Issue.1, pp.216-225, 2014.
- [3] Loughran, T. and McDonald, B., "When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks", *The Journal of Finance*, Vol.66, Issue.1, pp.35-65, 2011.
- [4] Vu, T., Le, H., and Pham, Q., "The predictive power of Vietnamese-language news sentiment on VN-Index returns", *Emerging Markets Review*, Vol.56, pp.101012, 2023.
- [5] Nguyen, T. and Pham, H., "Adapting VADER for Vietnamese financial news: A preliminary study", *Vietnam Journal of Computer Science*, Vol.11, Issue.1, pp.45-59, 2024.
- [6] Tran, P., Nguyen, L., and Hoang, D., "Short-term stock price prediction in Vietnam using XGBoost and LSTM models", *Journal of Asian Finance, Economics and Business*, Vol.11, Issue.3, pp.221-234, 2024.
- [7] Wang, J. and Vo, N., "Combining ARIMAX and sentiment analysis for HOSE stock forecasting", *Asian Economic Papers*, Vol.24, Issue.1, pp.78-95, 2025.
- [8] Chennupati, A., Prabhas, B., Aaditya, B., and Venugopalan, M., "Integrative Day Trading Stock Trend Prediction using Web Scraping and Sentiment Analysis", *Proceedings of 2022 IEEE 9th International Conference for Convergence in Technology (I2CT)*, pp.1-7, 2022.
- [9] Mitchell, R. and Chen, L., "Ethical web scraping in quantitative finance: A global survey of regulations", *Quantitative Finance*, Vol.22, Issue.8, pp.1401-1418, 2022.
- [10] Picault, M. and Renault, T., "Words vs. actions: News sentiment and the cross-section of stock returns", *Journal of Banking & Finance*, Vol.83, pp.1-20, 2017.
- [11] Smalović, J., Grčar, M., Lavrač, N., and Žnidaršič, M., "Stream-based active learning for sentiment analysis in the financial domain", *Information Sciences*, Vol.285, pp.181-203, 2014.
- [12] Oliveira, N., Cortez, P., and Areal, N., "Stock market sentiment lexicon acquisition using microblogging data and statistical measures", *Decision Support Systems*, Vol.85, pp.62-73, 2017.
- [13] Kraaijeveld, O. and De Smedt, J., "The predictive power of public Twitter sentiment for forecasting cryptocurrency prices", *Journal of International Financial Markets, Institutions and Money*, Vol.65, pp.101188, 2020.
- [14] Li, Q. and Pan, J., "Stock prediction based on deep learning and sentiment analysis", *Applied Soft Computing*, Vol.119, pp.109539, 2022.
- [15] Pagallo, U. and Ciani, J., "Anatomy of Web Data Scraping: Ethics, Standards, and the Troubles of the Law", *SSRN Electronic Journal*, 2023.
- [16] Zhang, W., Xu, H., and Feng, W., "A novel method for stock market prediction based on emotion analysis of online comments", *Proceedings of 2011 International Conference on Information*

*Management, Innovation Management and Industrial Engineering*, pp.397-401, 2011.

- [17] M.T.Stow "Hybrid Deep Learning Approach for Predictive Maintenance of Industrial Machinery using Convolutional LSTM Networks", *International Journal of Computer Sciences and Engineering*, Vol.12, Issue.4, pp.01-11, April 2024.
- [18] Santosh Kumar Miri, "Development of CBCS Evaluation Model by using Machine Learning Technique", *International Journal of Computer Sciences and Engineering*, Vol.12, Issue.4, pp.32-38, April 2024.

## AUTHORS PROFILE

**Ms. Tuyet Nhi Nguyen** is currently a first-year undergraduate student pursuing a Bachelor of Science in Artificial Intelligence at the University of Greenwich (Vietnam Campus), Ho Chi Minh City, Vietnam. This work constitutes her first scholarly publication and is based on an independent research project completed during her inaugural year in the AI Degree Programme. Her current research interests focus on the application of interpretable machine learning models to financial market forecasting, with particular emphasis on sentiment-augmented prediction in frontier markets such as the Ho Chi Minh Stock Exchange (HOSE).



**Ms. Sharmila Mathivanan** earned her Master's degree in Engineering, specializing in Information Technology, from K.S. Rangasamy College of Technology, which is affiliated with Anna University Chennai, Tamil Nadu, India, in the year 2014. With over 10 years of Teaching experience in the Department of Information Technology at various Engineering Colleges in India, she has been continuing her journey at University of Greenwich Vietnam since 2022. She has published 23 International research papers, holds two Patents related to Artificial Intelligence, and has one Copyright for a software application developed for the Transport Department of the Government of Tamil Nadu, India. Additionally, she received funding from the government for the successful completion of a project on "Predicting Mental depression through an Android application". Her teaching philosophy emphasizes critical thinking, creativity, and collaborative research in areas such as Artificial Intelligence, deep learning, web mining, medical diagnosis prediction, and the promotion of energy-efficient Artificial Intelligence applications.

